

Exercis 9 DAT 110

Exercise 9

Task 1

```
In [3]: 1 from scipy.stats import t
2
3 def calculate_critical_t(n, CL):
4     df = n-1
5
6     critical_t = t.ppf(1-(1-CL)/2, df)
7     return critical_t
8
9 A = calculate_critical_t(6, 0.9)
10 B = calculate_critical_t(21, 0.98)
11 C = calculate_critical_t(29, 0.95)
12 D = calculate_critical_t(12, 0.99)
13
14 print(f'A: {A}, B: {B}, C: {C}, D: {D}')
```

A: 2.015048372669157, B: 2.527977002740546, C: 2.048407141795244, D: 3.1058065132211006

Task 2

```
In [2]: 1 def find_p(n, T):
2     df = n - 1
3
4     return 2 * t.sf(abs(T), df)
5
6 A = find_p(11, 1.91)
7 B = find_p(17, -3.45)
8 C = find_p(7, 0.83)
9 D = find_p(28, 2.13)
10
11 print(f'A: {A}, B: {B}, C: {C}, D: {D}')
```

A: 0.08520487664515644, B: 0.0032935714037969134, C: 0.43830840156598644, D: 0.0424353859219

③ First I used the scipy t.ppf method to find the critical t. = -2,093

$$\bar{X} = \mu \pm t_{\text{critical}} \times \frac{s}{\sqrt{n}}$$

$$\bar{X} = 60 \pm (-2,093) \times \frac{8}{\sqrt{20}}$$

lower value = 56,26

upper value = 63,74

The sample mean has to be between these values or will be rejected: